# IFT6163: Project Proposal
# LoCoMotive : Comparing Behavior Cloning and Reinforcement Learning for Real World Robot Navigation

Byungsuk Min
Jad Bikarbass Azmi
Yanis Chikhar

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**Abstract**

Behavior Cloning vs Reinforcement Learning for Real-World Navigation. In this project, we study how different learning paradigms perform for real-world robot navigation under limited data. Using a LoCoBot mobile robot, we train goal-conditioned navigation policies that map visual observations to base velocity commands. We compare behavior cloning trained from teleoperated demonstrations collected on the real robot with reinforcement learning policies trained in simulation and transferred to the real robot. All methods use the same observation and action spaces to ensure a fair comparison. We evaluate performance on the real robot in indoor environments using success rate, collision rate, and path efficiency, and analyze failure modes to understand the strengths and weaknesses of each approach in practical robotic settings.

## 1 [4 pts] Project Method, How will the method work (2-3 paragraphs + figure(s)/math)

We study a goal conditioned indoor navigation task using a LoCoBot mobile platform in a structured lab environment. The robot must navigate from randomized start poses to predefined goal locations without access to a global map. At each timestep, the policy receives visual observations from the onboard RGB-D camera together with odometry and relative goal information, and outputs continuous linear and angular velocity commands. Episodes terminate when the robot reaches the goal region, collides, or exceeds a time limit. Performance will be evaluated exclusively on the physical robot using success rate, collision frequency, path length, and time to goal.

We compare two learning paradigms under a shared observation and action interface. First, a Behavior Cloning policy is trained from teleoperated demonstrations collected in the

real lab environment. Human operators guide the robot to goal locations while recording observations and control commands, and the policy is trained via supervised learning to imitate these trajectories. Second, a Reinforcement Learning policy is trained in simulation using a reward that encourages goal reaching and penalizes collisions and inefficient motion. The simulation environment, implemented in Gazebo or Habitat, will approximate the geometry and constraints of the real lab while matching control limits and observation formats. After convergence in simulation, the RL policy will be deployed directly on the real robot.

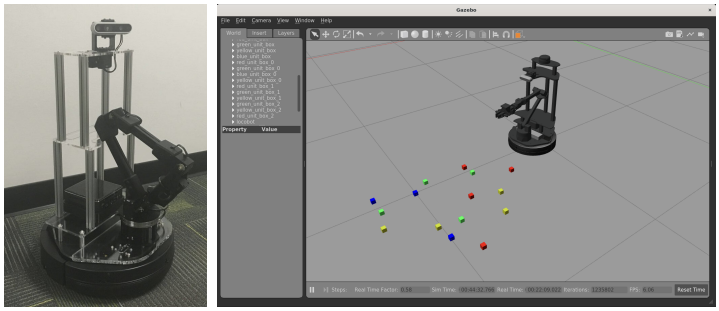

Figure 1: Left: Illustration of the LoCoBot mobile manipulator platform (image source: http://www.locobot.org/img/locobot.jpg). Right: Example Gazebo simulation environment for LoCoBot (image source: https://github.com/jacobazoulay/locobot/raw/main/assets/gazebo-locobot.png). We included these images for illustrative purposes. In the final report, we plan on replace them with images of our real robot and screenshots of the actual simulation environment used in our experiments.

Both policies share the same network architecture, action space, and evaluation protocol to ensure a fair comparison. Final experiments will be conducted under identical start and goal configurations on the physical robot. This setup enables a controlled study of data efficiency, robustness, and sim to real transfer in practical robotic navigation. If time permits and navigation performance is stable, we may extend the system with a simple scripted manipulation action triggered after successful goal reaching.

# 2 [2 pts] What is the related work? Provide references: (1 paragraph)

Imitation learning and reinforcement learning are two dominant paradigms for robotic policy learning. Behavior Cloning (BC) formulates policy learning as supervised regression from expert demonstrations and has been applied to autonomous navigation since early work in neural network-based driving Pomerleau [1989]. However, BC suffers from covariate shift and compounding errors when the learned policy encounters states outside the demonstration distribution, a limitation formally analyzed by DAgger Ross et al. [2011]. More recent imitation learning methods such as Generative Adversarial Imitation Learning (GAIL) Ho and Ermon [2016] and Behavioral Cloning from Observation (BCO) Torabi et al. [2018] aim to improve robustness and generalization in sequential decision making tasks. In contrast, Reinforcement Learning (RL) methods such as Proximal Policy Optimization (PPO) Schulman et al. [2017] optimize long term reward through interaction and have demonstrated strong performance in continuous control problems. For robotic navigation, simulation platforms such as Habitat Savva et al. [2019] enable scalable training, while sim-to-real transfer techniques including domain randomization Tobin et al. [2017], Peng et al. [2018] help bridge the gap between simulation and physical deployment. Our project builds upon these foundations by conducting a controlled empirical comparison of BC and RL for real world LoCoBot navigation under limited real data conditions, focusing on robustness and data efficiency.

# 3 [2 pts] What resources will you use (e.g. software) or need (e.g. hardware) and why: (1 paragraph)

We will use a LoCoBot mobile robot platform equipped with an RGB-D camera for real world data collection and evaluation. Teleoperation will be implemented through ROS to record demonstrations for Behavior Cloning. For simulation based Reinforcement Learning training, we will use Gazebo or Habitat to construct an indoor environment that approximates the lab layout while matching observation and control interfaces. Learning models will be implemented in PyTorch and trained using available GPU resources to accelerate training. All final evaluations will be conducted on the physical robot to ensure practical validation of the learned policies.

# 4 [2 pts] What new skills you(s) will learn from this project?

Through this project, we will gain practical experience in deploying learning based control policies on a real mobile robot. We will develop skills in teleoperation data collection, dataset construction, and supervised imitation learning for robotic control. In addition, we will learn how to design reward functions, train reinforcement learning policies in simulation, and analyze sim to real transfer performance. The project will also strengthen our ability to build reproducible experimental pipelines, evaluate learning systems under real world constraints, and perform systematic failure analysis on physical robotic platforms. If time permits, we will extend the system with a simple manipulation component using the robot arm after successful navigation. This will allow us to gain additional experience in coordinating perception, navigation, and manipulation within a unified control pipeline.

# 5 [2 pts] What is the timeline for the project? At least month-by-month granularity, ending with a complete project (8-12 pages with figures).

**Feb: Task definition and pipeline setup**

- Finalize navigation task in the lab environment, including start and goal configurations and evaluation metrics.
- Set up teleoperation and logging pipeline on the real robot.
- Collect initial demonstration dataset for Behavior Cloning.
- Construct the simulation environment in Gazebo or Habitat with matched observation and action interfaces.
- Implement baseline policy architecture shared by BC and RL.

**March: Model training and internal validation**

- Train and refine the Behavior Cloning policy using collected demonstrations.
- Train the Reinforcement Learning policy in simulation.
- Tune reward design and hyperparameters for stable RL training.
- Perform initial deployment tests of both policies on the real robot.
- Debug transfer issues and ensure consistent evaluation protocol.

**April: Real world evaluation and extension**

- Conduct controlled real world evaluation of both methods under identical start and goal conditions.
- Collect quantitative metrics including success rate, collision frequency, and path efficiency.
- Perform systematic comparison and failure analysis.
- If time permits, integrate a simple manipulation action using the robot arm after successful navigation.
- Prepare final report, visualizations, and presentation materials.

# References

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